

Assessing Logical Inference Capabilities of Large Language Models through RDF Schema Entailment Rules: A Multi-Level Evaluation

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Background / Motivation

- LLMs show strong NLP performance, but logical inference remains challenging
- LLMs rely on pretrained knowledge rather than explicit logical inference [3]
- Prior RDFS evaluation [1] lacked systematic rigor and covered only simple rules
- Counterfactual methods [2,3] show promise but are limited for RDFS inference
- We propose a benchmark using RDFS entailment rules with real-world and RDFS-specific counterfactual datasets to evaluate multi-rule application and rule selection quantitatively

Proposed Benchmark

RDFS Inference Tasks:

- 6 single-rule tasks using RDFS entailment rules
- 13 multi-rule tasks (7 two-rule + 6 three-rule combinations)

Rule	Premise	Conclusion
rdfs2	<i, rdfs:domain, X>, <a, i, b>	<a, rdf:type, X>
rdfs3	<i, rdfs:range, X>, <a, i, b>	<b, rdf:type, X>
rdfs5	<i, rdfs:subPropertyOf, j>, <j, rdfs:subPropertyOf, k>	<i, rdfs:subPropertyOf, k>
rdfs7	<i, rdfs:subPropertyOf, j>, <a, i, b>	<a, j, b>
rdfs9	<X, rdfs:subClassOf, Y>, <a, rdf:type, X>	<a, rdf:type, Y>
rdfs11	<X, rdfs:subClassOf, Y>, <Y, rdfs:subClassOf, Z>	<X, rdfs:subClassOf, Z>

Datasets:

- Real-world (RK): DBpedia, Wikidata, Linked Open Vocabularies
- Counterfactual (8 variants): S, NA/NR, SNA/SNR, GS, GSC, RND
- Random symbolic (NS): 8-character alphanumeric strings

Item	Content
rule	if <i, rdfs:domain, X> and <a, i, b> then <a, rdf:type, X>
premise_knowledge	<birthPlace, rdfs:domain, Animal>, <Cam_Woods, birthPlace, Montreal>
expected_output	<Cam_Woods, rdf:type, Animal>

CF Dataset	Description
S	Swaps subjects/objects, domains/ranges, reorders hierarchies
NA/NR	Adds "not" prefix to all/random classes and properties
SNA/SNR	Combines S with NA/NR modifications
GS	Shuffles resources across all types
GSC	GS + format names based on pre-shuffle position type
RND	Assigns random DBpedia vocabularies by type [3]

Task Levels:

- Inference strictly from given premises and rules only
 - Lv1: Apply single composite rule (e.g., rdfs2_3_7)
 - Lv2: Apply all given individual rules (e.g., rdfs2, rdfs3, rdfs7)
 - Lv3: Select and apply relevant rules from 6-rule set

Rule	Premise	Conclusion
rdfs2_3_7	<i, rdfs:domain, X>, <i, rdfs:range, Y>, <i, rdfs:subPropertyOf, j>, <a, i, b>	<a, rdf:type, X>, <b, rdf:type, Y>, <a, j, b>

Evaluation:

- Ground truth: Apache Jena RDFS reasoner
- Matching method: String similarity (triples), Exact match (rules)
- Metrics: Precision, Recall, F1 score

Evaluation Setup / Results

- 400 samples per dataset (except RND, NS: 100 samples)

Rule Application Performance (Average F1 Scores for each Rule Type)								
Rule Type	Task Level	Model	Dataset Type					
			RK	SNA	GS	GSC	RND	NS
1-rule	Lv1	GPT-4o mini	0.89	0.88	0.66	0.77	0.87	0.86
		GPT-4o	0.95	0.95	0.88	0.91	0.94	0.97
1-rule	Lv3	GPT-4o mini	0.70	0.71	0.49	0.56	0.65	0.77
		GPT-4o	0.94	0.93	0.83	0.91	0.91	0.91
3-rule	Lv2	GPT-4o mini	0.72	0.67	0.40	0.49	0.68	0.63
		GPT-4o	0.94	0.91	0.79	0.88	0.94	0.96
3-rule	Lv3	GPT-4o mini	0.64	0.56	0.30	0.38	0.58	0.56
		GPT-4o	0.89	0.86	0.71	0.82	0.88	0.88

Rule Selection Performance (Average F1 Scores for each Rule Type)							
Rule Type	Model	Dataset Type					
		RK	SNA	GS	GSC	RND	NS
1-rule	GPT-4o mini	0.84	0.85	0.76	0.77	0.85	0.80
	GPT-4o	0.98	1.00	0.98	0.99	1.00	0.99
2-rule	GPT-4o mini	0.89	0.90	0.81	0.82	0.95	0.96
	GPT-4o	0.99	0.99	0.89	0.91	0.99	1.00
3-rule	GPT-4o mini	0.82	0.82	0.84	0.85	0.87	0.91
	GPT-4o	0.96	0.96	0.93	0.96	0.97	0.98

Insights / Key Findings

LLMs Rely on Linguistic Cues, Not Pure Logic:

- Performance drops when resource names are shuffled (GS)
- Recovers when naming conventions are preserved (GSC)
- > Models use naming patterns as inference cues
- GPT-4o reduces this dependence compared to GPT-4o mini

Hallucination in Simple Tasks:

- GPT-4o mini applies unnecessary rules in 1-rule tasks
- Prioritizes semantic plausibility over strict logic
- GPT-4o better adheres to task constraints

Conclusion & Future Work

Conclusion:

- Proposed benchmark for evaluating LLM inference on RDFS
- LLMs rely on linguistic cues rather than pure logic
- Increasing model scale leads to more logical inference behavior
- Knowledge supplementation useful for incomplete KGs

Future Work:

- Evaluate reasoning-specialized models
- Systematic error analysis
- Extend evaluation to OWL

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References:

- [1] Wu et al. (2023). "Do PLMs Know and Understand Ontological Knowledge?" ACL
- [2] Ozeki et al. (2024). "NeuBAROCO: A Japanese Dataset for Evaluation of Syllogistic Reasoning Ability of Language Models" ANLP
- [3] Morishita et al. (2024). "JFLD: A Japanese Benchmark for Deductive Reasoning Based on Formal Logic" LREC-COLING



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